

TrackerBench: Stress-Testing Released Humanoid Motion-Tracking Policies

Shubham Singh

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Abstract

Whole-body motion-tracking policies for humanoid robots are evaluated almost exclusively by nominal tracking precision. We introduce TrackerBench, a stress-degradation benchmark that evaluates *released* tracking checkpoints, each on its authors’ own simulation deploy stack, under four seeded perturbation axes: external pushes, torso payload, observation noise, and action latency. Perturbation realizations are identical across policies by construction, failure is scored under two termination rules, and degradation is measured against each policy’s own unperturbed baseline. Across 1,700 paired rollouts on the Unitree G1 we find that nominal precision does not predict robustness: the policy that tracks best on all eight test motions is out-survived, significantly and stably, on 33 stress conditions by a nominally worse policy; a per-motion specialist tracks extreme motions at 2.3 cm yet collapses under pushes an order of magnitude weaker than generalists tolerate; and one “generalist” policy fails on ordinary walking motions produced by a different retargeting pipeline before any perturbation is applied. Each policy exhibits a distinct vulnerability fingerprint that no single scalar metric captures.

1. Introduction

Humanoid whole-body tracking policies — GMT, TWIST, SONIC, BeyondMimic, and their successors — are compared by how precisely they reproduce reference motions under ideal conditions, typically mean per-joint position error (MPJPE) and a success rate. Deployment, however, is dominated by disturbances: contact impulses, payload variation, sensor noise, and computation delays. No existing evaluation measures how released tracking policies *degrade* under such disturbances, and robustness claims in the literature are supported only by each paper’s own demonstrations.

TrackerBench fills this gap with three design commitments:

1. **Evaluate released checkpoints, not reimplementations.** Each policy runs on its authors’ own published simulation deploy stack (model file, PD gains, observation pipeline), reproduced verbatim; the policy remains a black box.
2. **Paired perturbations.** All stochastic perturbation realizations are a deterministic function of the evaluation cell, never of the policy — every policy faces the same shoves and the same noise, enabling paired statistics.
3. **Protocol-artifact controls.** Every headline quantity must survive two different failure definitions and normalization to each policy’s own unperturbed baseline, so that differences in re-hosting fidelity cannot masquerade as robustness differences.

2. Benchmark

2.1 Setup

All experiments use the Unitree G1 humanoid in MuJoCo (CPU). The motion set comprises the eight retargeted mocap clips released with GMT: walking, walk-to-stand, squat, crouch-walk, two dances, a kick while walking, and an airborne kick. A rollout runs at each policy’s native control rate (50 Hz for all families here) for the clip duration, capped at $T = 30$ s; the first $t_0 = 0.75$ s is a settling transient excluded from scoring.

2.2 Methods compared

Policy	Type	Format	Act. DoF	Nominal MPJPE (cm)
GMT (arXiv 2506.14770)	generalist	TorchScript	23	1.39 – 2.85
TWIST (CoRL 2025)	generalist (teleop)	TorchScript	23	1.67 – 4.12
PBHC / KungfuBot (NeurIPS 2025)	per-motion specialist	ONNX $\$ \times \3	23	2.28 – 2.63 †
OpenTrack (arXiv 2509.13833)	generalist	ONNX	29	excluded (§4.4)

† evaluated on its own three released extreme-motion clips (specialist track); the remaining families share the eight-clip set. Integration of SONIC, Humanoid-GPT, and Any2Track is ongoing.

2.3 Perturbation axes

Each cell of the benchmark is a tuple (clip, axis, severity, seed) with three seeds per cell. The random draws in a cell derive from a hash of the cell identifier alone, so **every policy receives identical perturbation realizations**. The axes and severity grids are:

Pushes. A horizontal force of magnitude F is applied at the pelvis in the world frame for 0.2 s every 2.0 s beginning at $t = 1$ s, with per-event direction $\theta_k \sim \mathcal{U}(0, 2\pi)$:

$$\mathbf{f}(t) = F [\cos \theta_k, \sin \theta_k, 0]^\top, \quad t \in [t_k, t_k + 0.2 \text{ s}],$$

with $F \in \{50, 100, 150, 200, 300, 400\}$ N (a 0.2 s, 200 N pulse imparts a 40 N · s impulse, about 1.1 m/s of pelvis velocity change on the ~35 kg G1).

Payload. A mass $m \in \{2, 4, 5, 8, 10, 12, 16, 24\}$ kg is added to the torso link, with the body’s diagonal inertia scaled consistently: $m' = m_0 + m$, $\mathbf{I}' = \mathbf{I}(1 + m/m_0)$.

Observation noise. Zero-mean Gaussian noise is added to the *policy’s* proprioceptive inputs (the simulator state is untouched) at every control step:

$$\tilde{o}_c = o_c + \varepsilon_c, \quad \varepsilon_c \sim \mathcal{N}(0, (\lambda \sigma_c)^2),$$

with base scales $\sigma = 0.1$ rad/s (gyro), 0.01 rad (orientation), 0.01 rad (joint position), 0.1 rad/s (joint velocity), and level $\lambda \in \{1, 2, 4, 8, 16\}$. For policies whose observation uses a gravity vector instead of Euler angles, the orientation noise is applied as a random small rotation of that vector.

Action latency. The actuation target computed at control step j is applied at step $j + k$, $k \in \{1, 2, 3, 5\}$ (20 – 100 ms at 50 Hz), implemented as a delay queue on the PD setpoint.

2.4 Metrics

Tracking error. Let $x_b^a(t)$ and $x_b^r(t)$ be the world positions of body b in the rollout and the reference, and $R^a(t), R^r(t)$ the corresponding pelvis rotations. The *root-frame* error

$$e(t) = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \left\| R^{a\top}(x_b^a - x_{\text{root}}^a) - R^{r\top}(x_b^r - x_{\text{root}}^r) \right\|_2$$

measures pose tracking independently of global drift and heading — necessary because several policies (e.g. GMT) are heading-free by construction: their observations contain no world yaw, so world-frame error grows without bound even under perfect local tracking. MPJPE is the time-average of $e(t)$ over the scored window.

Termination rules. Failure is scored post hoc under two rules from a single rollout:

$$t_A = \inf \{ t : z_{\text{pelvis}}(t) < 0.35 \text{ m} \} \quad (\text{fall}),$$

$$t_B = \min(t_A, \inf \{ t : e(t) > 0.35 \text{ m} \}) \quad (\text{fall or divergence}).$$

Requiring conclusions to hold under both rules guards against the failure definition itself driving the result.

Survival. The primary per-cell metric is the survival fraction

$$S = \text{clip} \left(\frac{t_{\text{fail}} - t_0}{T - t_0}, 0, 1 \right),$$

i.e. the fraction of the scored window completed before failure ($S=1$ if no failure occurs). Degradation is always reported relative to the same policy’s unperturbed baseline on the same clip, $\Delta S = S - S^{\text{nominal}}$, which removes re-hosting fidelity as a confound (all nominal S values here are 1.0).

Paired comparison and rank crossovers. For policies P, Q in cell c with paired seeds $s \in \{1, 2, 3\}$, define the per-seed gap $d_s = S_{c,s}^P - S_{c,s}^Q$. A cell exhibits a *rank crossover* if the policy with worse nominal MPJPE has

$$\text{median}_s d_s > 2 \hat{\sigma}(d_s)$$

and the inequality holds under **both** termination rules. Stability is quantified by seed test–retest: across all non-trivial cells, the fraction where $\text{sign}(d_1)$ agrees with $\text{sign}(\frac{1}{2}(d_2 + d_3))$.

3. Results

3.1 Nominal precision does not predict robustness

GMT is the more precise tracker on **all eight** clips (1.39 – 2.85 cm vs 1.67 – 4.12 cm). Under stress, the ordering reverses in 33 of 77 non-trivial cells — all 33 significant under the paired criterion

Stress-degradation curves — GMT vs TWIST, 8 shared clips, paired perturbations (rule A)

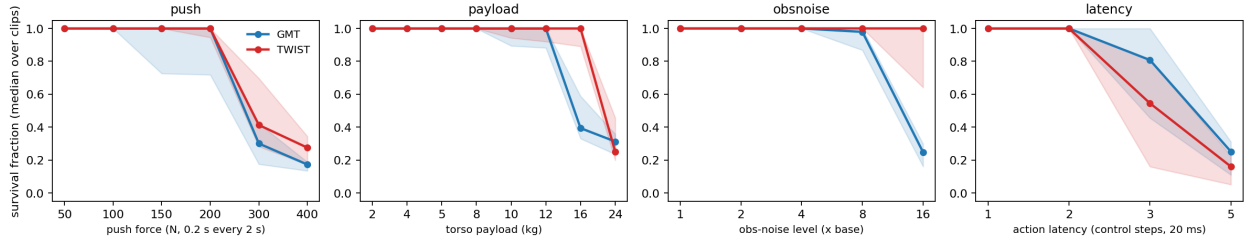


Figure 1: Stress-degradation curves for the two generalists on the eight shared clips (median over clips; bands: IQR; rule A). Payload and observation noise reverse the nominal ordering; latency preserves it.

Vulnerability fingerprints — three policy families, three failure geometries

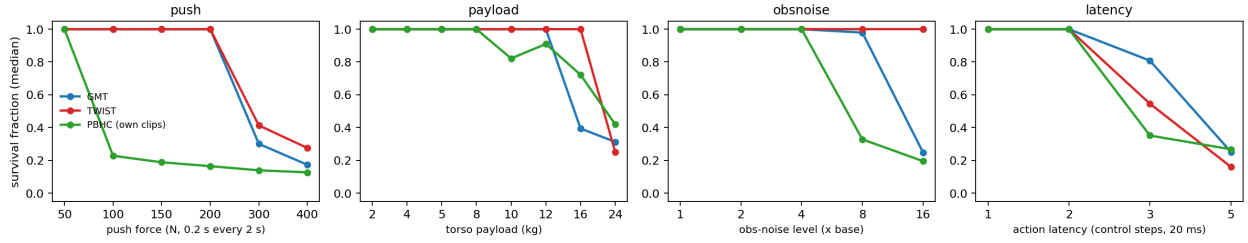


Figure 2: Vulnerability fingerprints across three families (PBHC on its own clips). Each family fails in a different region of the perturbation space.

and identical under both termination rules, with seed test–retest agreement 0.883. The largest effects: at 16 kg payload on basic walking GMT survives 15% of the clip while TWIST completes it ($\Delta S = 0.85$, seed variance zero); at $16\times$ observation noise TWIST walks through clips that terminate GMT within seconds. Fourteen reverse cells also exist — GMT out-survives TWIST under 60 ms latency and several push conditions — so neither policy dominates.

3.2 Vulnerability fingerprints are axis-specific

The natural summary of a policy is not a scalar but a *fingerprint*: the vector of severities at which its median survival crosses $\frac{1}{2}$ per axis. For GMT: 300 N (push), 16 kg (payload), $16\times$ (noise), 60–100 ms (latency). For TWIST: greater than 300 N, greater than 24 kg, greater than $16\times$, but 60 ms latency. For PBHC: **100 N**, ~ 24 kg, $8\times$, 60 ms. Aggregating these into one number would discard precisely the information a deployer needs.

3.3 Specialists trade robustness for precision

PBHC’s per-motion checkpoints track extreme kung-fu motions at 2.3 cm — yet their push tolerance collapses between 50 N and 100 N (survival $1.00 \rightarrow 0.23$), an order of magnitude below the generalists, and observation noise degrades them at $8\times$ where generalists remain clean. Payload tolerance, in contrast, is intact through 8 kg. A caveat applies in both directions: PBHC’s clips are quasi-static

stances with no gait momentum to absorb impulses, underscoring why a shared, neutral clip set is a benchmark requirement rather than a convenience.

3.4 “Generalist” scope is bounded by the retargeting pipeline

OpenTrack re-hosts flawlessly — our evaluation loop is bit-identical to the authors’ own environment (maximum state deviation 0.0 over 150 control steps), and it tracks the authors’ own LAFAN1 reference clips at 11.8 – 16.4 mm root-frame error over 20 s rollouts. On five of the eight shared clips, however, it falls within seconds *without any perturbation*. The shared clips contain ordinary walking and dancing on the same robot; they differ only in the retargeting pipeline that produced them. The policy’s generalization is therefore bounded by its training pipeline’s retargeting distribution — a failure mode invisible to any evaluation that uses each policy’s own reference files, and detectable only by a neutral cross-pipeline benchmark. Per the benchmark’s pre-registered admission rule (nominal success on at least six of eight shared clips), OpenTrack is excluded from head-to-head scoring and reported here as a finding.

4. Limitations

The head-to-head comparison currently covers two generalists; three further families (SONIC, Humanoid-GPT, Any2Track) are being integrated. All results are simulation-to-simulation; hardware validation is future work. The shared clips originate from GMT’s release, which if anything favors GMT — making the observed crossovers conservative — but a multi-pipeline neutral clip set is under construction. Three seeds per cell suffice for the paired sign statistics reported here; the public release will increase seed count and adopt survival-analysis aggregation (restricted mean survival time) with censoring handled explicitly.

5. Conclusion

Stress-testing released humanoid tracking policies under paired, seeded perturbations reveals structure that nominal metrics cannot: rank crossovers between precision and robustness, axis-specific vulnerability fingerprints, a specialist–generalist robustness trade-off, and retargeting-pipeline-bounded generalization. TrackerBench provides the protocol, controls, and metrics to measure this structure reproducibly, at CPU cost, for any released checkpoint.